

Lab 12**SQL**

In this lab session, we are going to interact with our simple movies' database using MySQL.

Our database keeps track of movies that different users have watched. The ER diagram is the following:

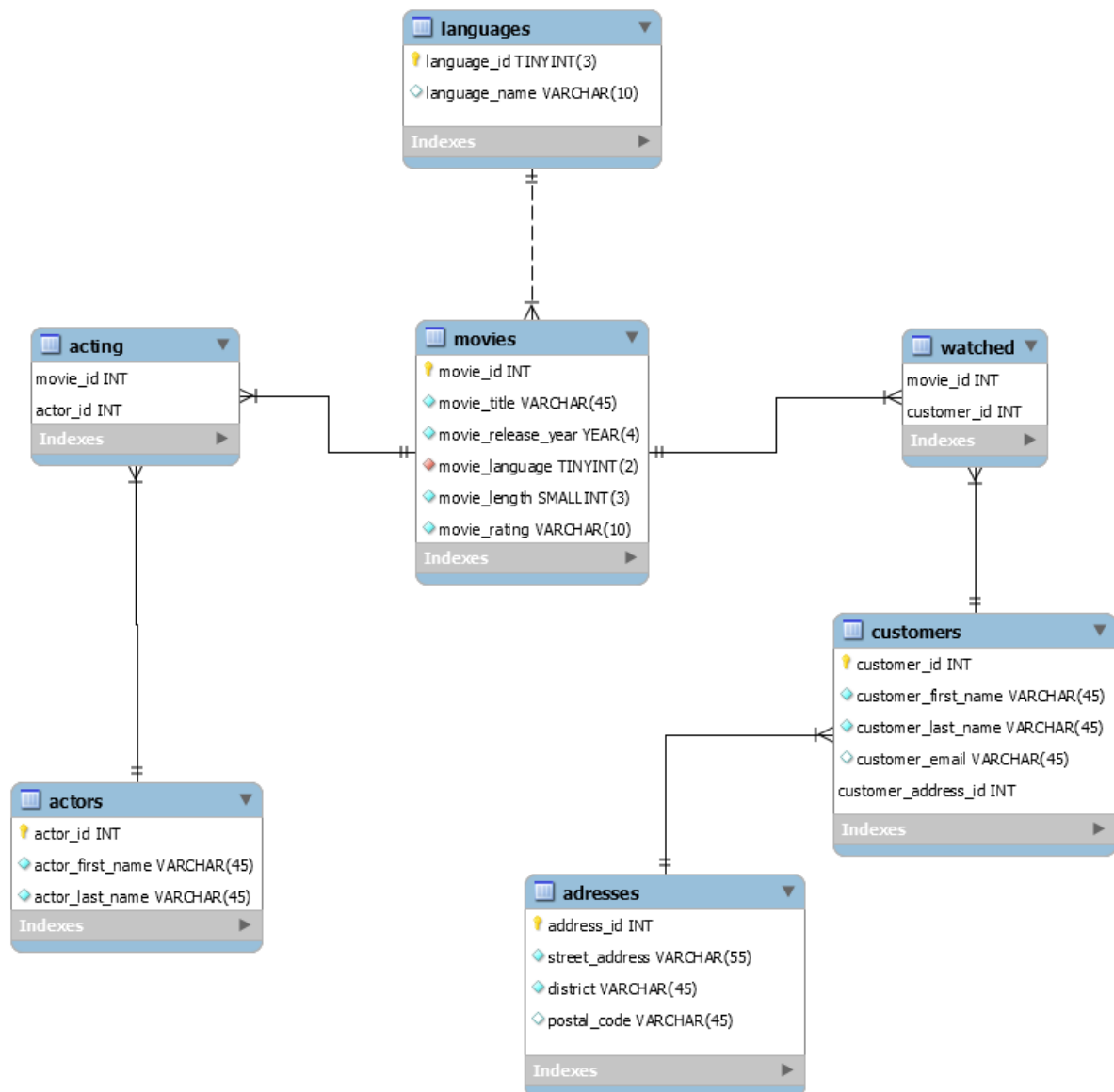


Figure 1 Movies ER Diagram (adaptation from sakila sample DB)

TASK 1 Explain what each of the following queries do.

--Query 1

```
SELECT movies.movie_title, languages.language_name FROM movies LEFT JOIN languages ON language_id=movie_language
```

--Query 2

```
SELECT movies.movie_title, languages.language_name FROM movies INNER JOIN languages ON languages.language_id=movies.language_id
```

--Query 3

```
SELECT COUNT(*) AS "movies with language defined" FROM movies LEFT JOIN languages ON languages.language_id=movies.movie_language;
```

--Query 4

```
SELECT movies.movie_title,movies.movie_length FROM movies
WHERE movies.movie_length=(
SELECT MAX(movies.movie_length) FROM movies)
```

--Query 5

```
SELECT movies.movie_title, actors.actor_last_name
FROM movies INNER JOIN movies_actors_table USING(movie_id)
INNER JOIN actors USING(actor_id)
WHERE actors.actor_last_name REGEXP "CHASE"
```

--Query 6

```
SELECT movies.movie_title FROM movies
INNER JOIN movies_actors_table USING(movie_id) INNER JOIN (
        SELECT actor_id FROM actors
        WHERE actors.actor_last_name ="CHASE") as T2
ON movies_actors_table.actor_id=T2.actor_id
```

TASK 2 QUERIES Perform the following queries:

1. Get a list of all customers whose last names start with M, F, R or T.
2. Get a list of all the customers whose last name is starts with R and whose email is a gmail account.
3. Get the title of the shortest movie(s).
4. Get movie titles for movies in French released after 2019.
5. Get movie titles and actors' name and surname for all movies with a PG rating.
6. Get most common language in movies between 1970 and 1990.
HINT: Use COUNT(*), ORDER BY and limit results to 1
7. Get most common language in R rated movies between 2000 and 2010.
8. Get the name and surname of the actor that has performed in the most movies.
9. Get the number of times watched per movie. Save this into a view.
10. Get 10 most watched movies using the view created in question 9.
11. Get 10 most watched movies using a subquery.
12. Get most watched movie per rating.
13. Get most watched movie per rating since 2015.

TASK 3 OPTIMIZE YOUR DATABASE

- A. Based on the expected use of the database, what indexes would you add to the schema?